1.(a) Normal-Inv-χ^2 的核心项为 $(V, \tau^2; \mu_0, \frac{\sigma^2}{k_0})$

$$(\sigma^2)^{-\frac{V+3}{2}} \exp\left[-\frac{V\tau^2+k_0(\mu-\mu_0)^2}{2\sigma^2}\right]$$

似然函数 $f(y|\vec{\theta}) \propto (\sigma^2)^{-\frac{n}{2}} \exp\left[-\frac{1}{2\sigma^2}[(n-1)s^2+n(\bar{y}-\mu)^2]\right]$

posterior $p(\vec{\theta}|y) \propto (\sigma^2)^{-\frac{n+V+3}{2}} \exp\left\{-\frac{1}{2\sigma^2}[V\tau^2+(n-1)s^2+n(\bar{y}-\mu)^2+k_0(\mu-\mu_0)^2]\right\}$

$\sim$ Normal-Inv-χ^2 $\left(n+V,\ \frac{V_0\tau^2+(n-1)s^2+\frac{k_0 n}{k_0+n}(\bar{y}-\mu_0)^2}{V_0+n};\ \frac{\mu_0 k_0+n\bar{y}}{k_0+n},\ \frac{\sigma^2}{k_0+n}\right)$

(b) Find $p(\sigma^2|\mu, y)$

$$p(\sigma^2, \mu|y) \propto (\sigma^2)^{-\frac{n+V+3}{2}} \exp\left\{-\frac{1}{2\sigma^2}[V\tau^2+(n-1)s^2+n(\bar{y}-\mu)^2+k_0(\mu-\mu_0)^2]\right\}$$

$$f(\sigma^2, \mu)$$

$$p(\mu|y) \propto \left[1+\frac{n(\mu-\bar{y})^2}{(n-1)s^2}\right]^{-\frac{n}{2}} \quad (\text{Student-}t_{n-1}(\bar{y}, \tfrac{s^2}{n}))$$

$$\therefore p(\sigma^2|\mu, y) \propto \left[1+\frac{n(\mu-\bar{y})^2}{(n-1)s^2}\right]^{-\frac{n}{2}} \cdot (\sigma^2)^{-\frac{n+V+3}{2}} \exp\left\{\frac{f(\sigma^2,\mu)}{}\right\}$$

(c) 1°: 直接应用先验样本量 $\to 0$:

令 $\sigma^2 \sim$ Inv-$\chi^2(V_0, \tau^2)$ 中的 $V \to 0$

$\mu|\sigma^2 \sim N(\mu_0, \frac{\sigma^2}{k_0})$ 中的 $k_0 \to 0$:

得到: $p(\sigma^2) \xrightarrow{V\to 0} (\sigma^2)^{-1}$

$p(\mu|\sigma^2) \xrightarrow{k_0\to 0} U(-\infty, +\infty) \propto 1$

2°: 若令 $p(\mu, \sigma^2)$ 含共轭先验 $\propto (\sigma^2)^{-1}$

得到: $-\frac{V^*+3}{2} = -1$, $V^* = -1$

$V^*\tau^{*2}+k_0^*(\mu-\mu_0)^2 = 0 \Rightarrow \tau^* = 0, k_0^* = 0$

从而 $p(\sigma^2) \xrightarrow{V\to V^*, \tau^2\to\tau^{*2}} (\sigma^2)^{-1}$

$p(\mu|\sigma^2) \xrightarrow{k_0\to k_0^*} \propto 1$

(d) $p(\tilde{y}|y) = \iint p(\tilde{y}|y,\mu,\sigma^2)\, p(\mu,\sigma^2|y)\, d\mu\, d\sigma^2$

$\tilde{y} \perp\!\!\!\perp y$ $\iint p(\tilde{y}|\mu,\sigma^2)\, p(\mu,\sigma^2|y)\, d\mu\, d\sigma^2$

$p(\tilde{y}|\mu,\sigma^2) = N(\mu,\sigma^2)$，与 $p(\mu,\sigma^2|y)$ 相乘后形式不变

仍为 Normal-Inv-χ^2 分布，参数 $n=1$，$\bar{y}=\tilde{y}$

将其代入更新公式（见(c)中推导）

$$\begin{cases} \tilde{v}_0 = 1+v \\ \tilde{\sigma}^2 = \dfrac{v_0\sigma^2 + \frac{k_0}{k_0+1}(\tilde{y}-\mu_0)^2}{v_0+1} \\ \tilde{\mu}_0 = \dfrac{k_0\mu_0+\tilde{y}}{k_0+1} \\ \tilde{k}_0 = 1+k_0 \end{cases}$$

2p可

$-(n+2)\sigma^2 + (n-1)s^2 + n(\bar{y}-\mu)^2 = 0$

(e) ① 抽取方差参数 σ^*

② 在固定 σ^* 的条件下抽取均值参数 μ^*

③ 在 $N(\mu^*, \sigma^{*2})$ 分布中抽取样本 $\tilde{y}^*$

2. $p(\sigma|y) \propto (\sigma^2)^{-1} \cdot (\sigma^2)^{-\frac{n}{2}} \exp\{[(n-1)s^2 + n(\bar{y}-\mu)^2]/-2\sigma^2\}$

$= (\sigma^2)^{-\frac{n+2}{2}} \exp\{[(n-1)s^2 + n(\bar{y}-\mu)^2]/-2\sigma^2\}$

$\log p(\sigma|y) = -(n+2)\log\sigma - \frac{(n-1)s^2+n(\bar{y}-\mu)^2}{2\sigma^2}$

$\frac{\partial \log(\cdot)}{\partial \sigma^*} = \frac{-n-2}{\sigma} + \frac{(n-1)s^2+n(\bar{y}-\mu)^2}{\sigma^3} \Rightarrow \hat{\sigma}^2 = \frac{(n-1)s^2+n(\bar{y}-\mu)^2}{n+2}$ $\hat{\mu}=\bar{y}$ $\frac{n-1}{n+2}$

$\frac{\partial \log(\cdot|y)}{\partial \mu} = \frac{n(\bar{y}-\mu)}{\sigma^2} \Rightarrow \hat{\mu} = \bar{y}$ $\frac{n+2}{\sigma^2} - \frac{3(n-1)s^2}{\sigma^4}$

$\hat{\theta} = (\bar{y}, \hat{\sigma}^2)$

$$I(\hat{\theta}) = \begin{bmatrix} \frac{n\hat{\sigma}^2 - 3(n-1)S^2}{\hat{\sigma}^4} & 0 \\ 0 & -\frac{n}{\hat{\sigma}^2} \end{bmatrix} = \begin{bmatrix} \frac{-2(n-1)S^2}{\hat{\sigma}^4} & 0 \\ 0 & -\frac{n}{\hat{\sigma}^2} \end{bmatrix}$$

$$p(\theta|y) \overset{n\to\infty}{\sim} N(\hat{\theta}, -I(\hat{\theta})^{-1})$$

$$= N\left(\begin{bmatrix} \bar{y} \\ \frac{n-1}{n+2}S^2 \end{bmatrix}, \begin{bmatrix} \frac{\hat{\sigma}^4}{2(n-1)S^2} & 0 \\ 0 & \frac{\hat{\sigma}^2}{n} \end{bmatrix} \right)$$

where $\hat{\sigma}^2 = \frac{n-1}{n+2}S^2$

3. $p(\lambda) = \frac{\lambda^k}{k!}e^{-\lambda}$

(a) $L(\lambda) = \prod_i \frac{\lambda^{k_i}}{k_i!}e^{-\lambda} = \prod_i \frac{1}{k_i!} \cdot \lambda^{\sum k_i} e^{-n\lambda}$ ... λ

取 log:

$\log L(\lambda) = \text{const.} + (\sum k_i)\log\lambda - n\lambda$

$\frac{\partial \log L}{\partial \lambda} = -n + \frac{\sum k_i}{\lambda}$

$\frac{\partial^2 \log L}{\partial \lambda^2} = -\frac{\sum k_i}{\lambda^2}$

$\mathbb{E}_k\left[-\frac{\partial^2 \log L}{\partial \lambda^2}\right] = \frac{1}{\lambda^2}\mathbb{E}[\sum k_i] = \frac{n\lambda}{\lambda^2} = \frac{n}{\lambda}$

$J(\lambda) = \sqrt{I(\lambda)} \propto \lambda^{-\frac{1}{2}}$

(b) $p(\lambda|y) \propto p(\lambda)\, p(y|\lambda)$

$= \lambda^{-\frac{1}{2}} \cdot \lambda^{\sum y_i} e^{-n\lambda} = \lambda^{\sum y_i - \frac{1}{2}} e^{-n\lambda}$

(c) $p(X_1, X_2, \cdots, X_n|\lambda) = \prod_{i=1}^{n} \frac{1}{X_i!} \lambda^{\sum X_i} e^{-n\lambda}$

$p(X_1, \cdots X_n) = \int_0^{+\infty} p(X_1, \cdots X_n|\lambda)\, d\lambda$

$= \prod \frac{1}{X_i!} \lambda^{\sum X_i} \cdot \frac{1}{n}$

$\int_0^{+\infty} e^{-n\lambda} d\lambda$

$\frac{1}{n} - e^{-n\lambda}\Big|_0^{+\infty}$

(d) $\log p(\lambda|y) = (\sum y_i - \frac{1}{2}) \log \lambda - n\lambda$

$\frac{\partial \log p(\lambda|y)}{\partial \lambda} = (\sum y_i - \frac{1}{2})/\lambda - n \Rightarrow \hat{\lambda} = \frac{1}{n}(\sum y_i - \frac{1}{2})$

$\frac{\partial^2 \log p(\lambda|y)}{\partial \lambda^2} = -\frac{\sum y_i - \frac{1}{2}}{\lambda^2} \Rightarrow \lambda = \hat{\lambda}$ 时 $\frac{n^2}{(\sum y_i - \frac{1}{2})}$

$\therefore p(\lambda|y) \sim N\left(\frac{1}{n}(\sum y_i - \frac{1}{2}), \frac{\sum y_i - \frac{1}{2}}{n^2}\right)$

4. 因为 $n \to \infty$ 时, $Var(\cdot) \to 0$. ($\cdot$ 表任一随机变量), 也即所有 pdf 会收敛到一个确定点上, 在该点处, $f(\theta_0) = \phi_0$ 成立.

5.